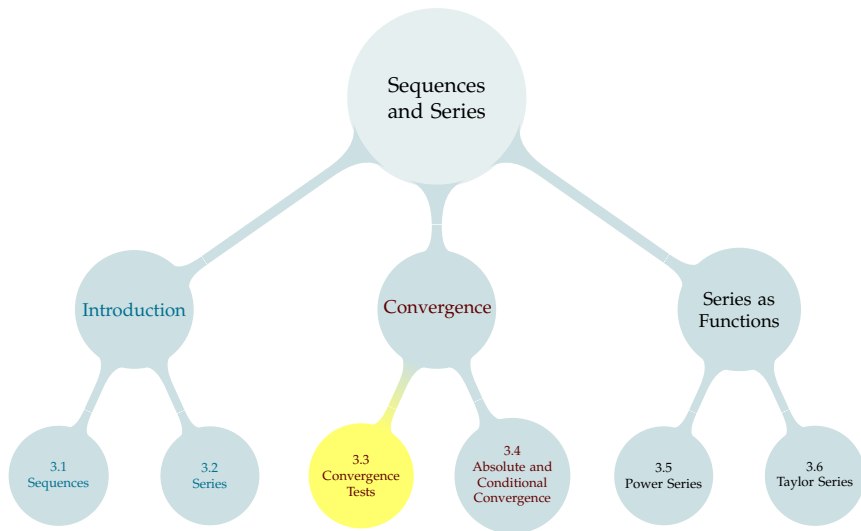


TABLE OF CONTENTS



For a convergent geometric or telescoping series, we can easily determine what the series converges *to*.

For other types of series, finding out what the series converges to can be very difficult. It is often necessary to resort to approximating the full sum by, for example, using a computer to find the sum of the first N terms, for some large N . But before we even try to do that, we should at least know *whether or not the series converges*.

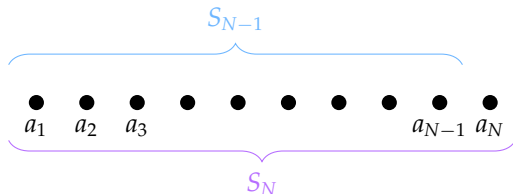
Suppose a series $\sum_{n=1}^{\infty} a_n$ converges to a limit L . Let $S_N = \sum_{n=1}^N a_n$.

$$\lim_{N \rightarrow \infty} S_N =$$

$$\lim_{N \rightarrow \infty} S_{N-1} =$$

$$\lim_{N \rightarrow \infty} [S_N - S_{N-1}] =$$

$$\lim_{N \rightarrow \infty} a_N =$$



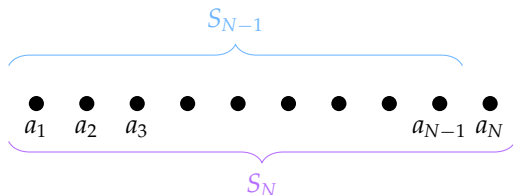
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Every convergent series has its N^{th} term, a_N , tending to 0 as $N \rightarrow \infty$.

Divergence Test

If the sequence $\{a_n\}_{n=c}^{\infty}$ fails to converge to zero as $n \rightarrow \infty$, then the series $\sum_{n=c}^{\infty} a_n$ diverges.

Divergence Test

If the sequence $\{a_n\}_{n=c}^{\infty}$ fails to converge to zero as $n \rightarrow \infty$, then the series $\sum_{n=c}^{\infty} a_n$ diverges.

Do the following series diverge?

▶ $\sum_{n=0}^{\infty} (-1)^n$

▶ $\sum_{n=10}^{\infty} \left(\frac{1}{10} + \frac{1}{2^n} \right)$

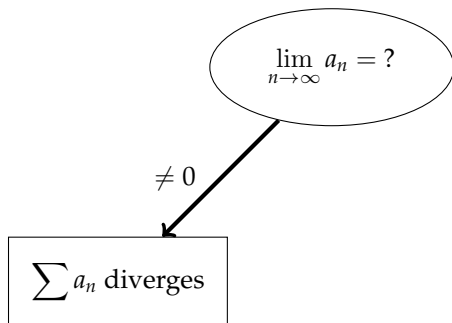
▶ $\sum_{n=15}^{\infty} \frac{e^n}{2e^n - 1}$

▶ $\sum_{n=15}^{\infty} \frac{1}{n}$

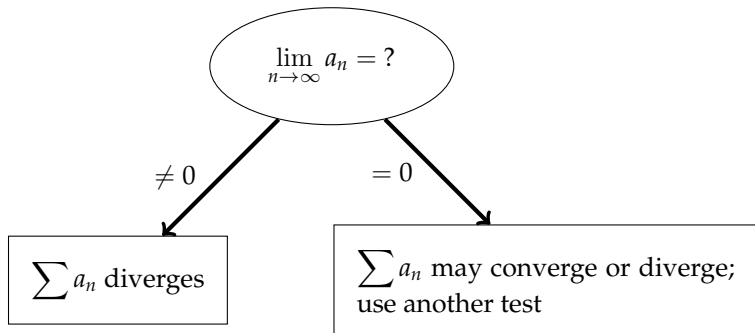
USING THE DIVERGENCE TEST FOR $\sum a_n$

$$\lim_{n \rightarrow \infty} a_n = ?$$

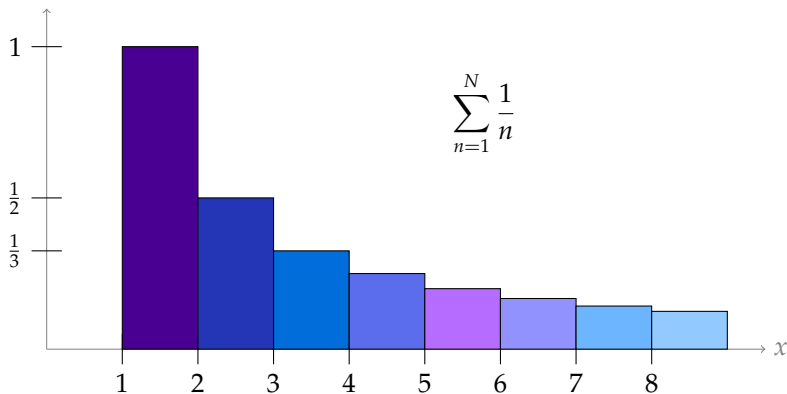
USING THE DIVERGENCE TEST FOR $\sum a_n$



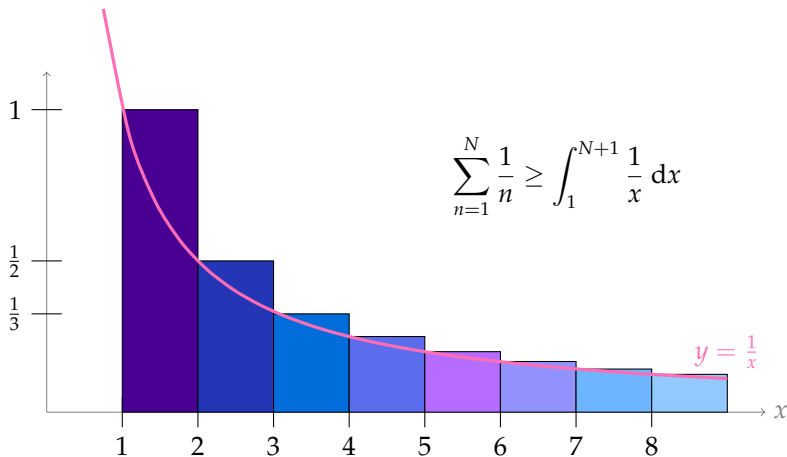
USING THE DIVERGENCE TEST FOR $\sum a_n$



HARMONIC SERIES: $\sum_{n=1}^{\infty} \frac{1}{n}$



HARMONIC SERIES: $\sum_{n=1}^{\infty} \frac{1}{n}$



$$\sum_{n=1}^{\infty} \frac{1}{n}$$

DIVERGES



1



$\frac{1}{2}$



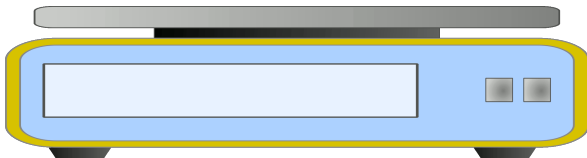
$\frac{1}{3}$



$\frac{1}{4}$



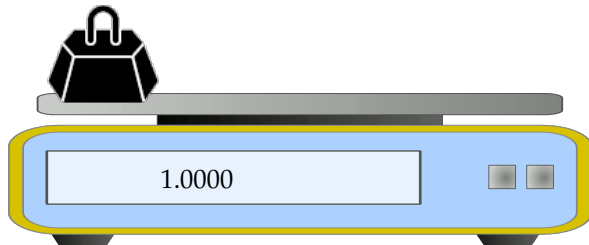
$\frac{1}{5}$



$$\sum_{n=1}^{\infty} \frac{1}{n}$$

DIVERGES

$$S_1 = 1.0000$$

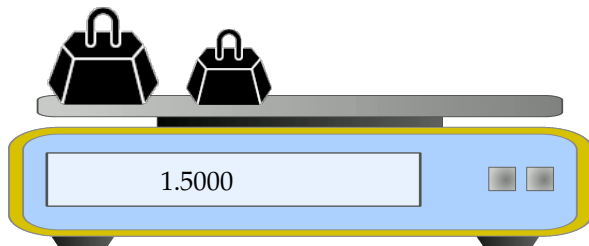
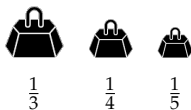


$$\sum_{n=1}^{\infty} \frac{1}{n}$$

DIVERGES

$$S_1 = 1.0000$$

$$S_2 = 1.5000$$



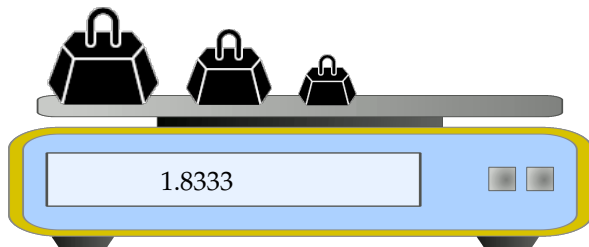
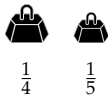
$$\sum_{n=1}^{\infty} \frac{1}{n}$$

DIVERGES

$$S_1 = 1.0000$$

$$S_2 = 1.5000$$

$$S_3 = 1.8333$$



$$\sum_{n=1}^{\infty} \frac{1}{n}$$

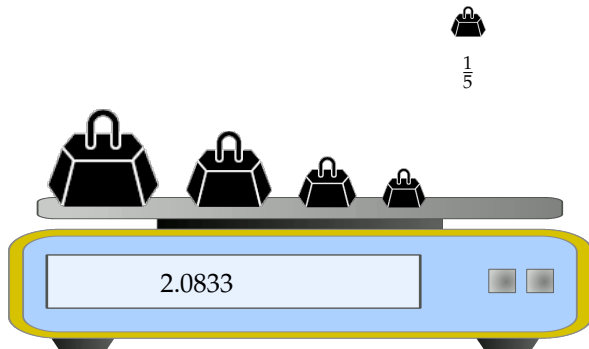
DIVERGES

$$S_1 = 1.0000$$

$$S_2 = 1.5000$$

$$S_3 = 1.8333$$

$$S_4 = 2.0833$$



$$\sum_{n=1}^{\infty} \frac{1}{n}$$

DIVERGES

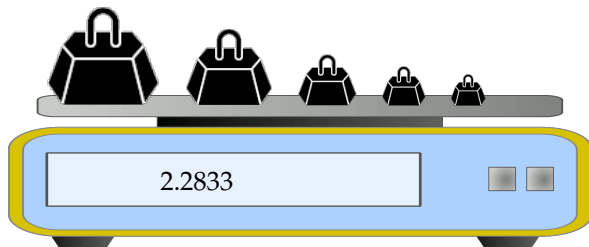
$$S_1 = 1.0000$$

$$S_2 = 1.5000$$

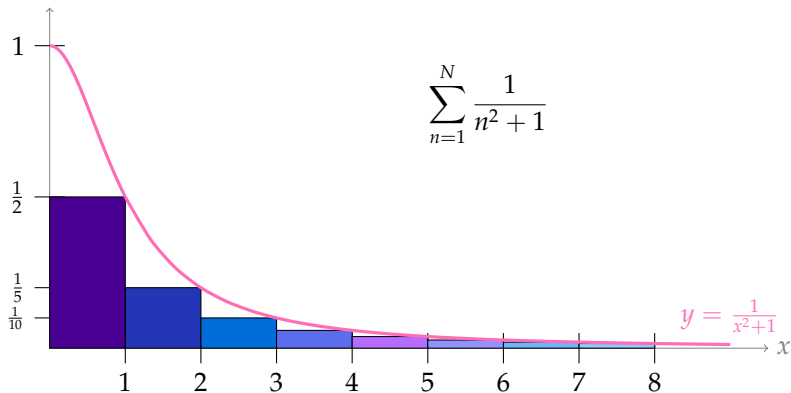
$$S_3 = 1.8333$$

$$S_4 = 2.0833$$

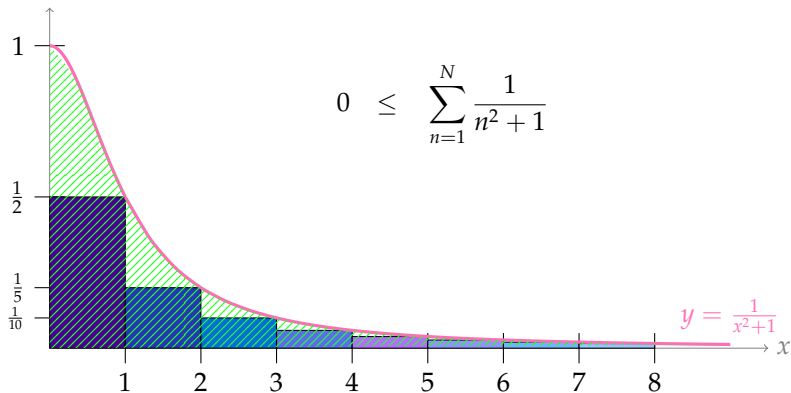
$$S_5 = 2.2833$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

CONVERGES



$$\frac{1}{2}$$



$$\frac{1}{5}$$



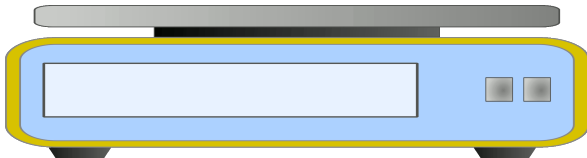
$$\frac{1}{10}$$



$$\frac{1}{17}$$



$$\frac{1}{26}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

CONVERGES

$$S_1 = 0.5000$$



$$\frac{1}{5}$$



$$\frac{1}{10}$$



$$\frac{1}{17}$$



$$\frac{1}{26}$$



0.5000

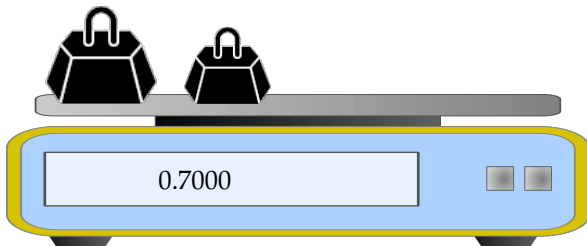
$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

CONVERGES

$$S_1 = 0.5000$$

$$S_2 = 0.7000$$


$$\frac{1}{10} \quad \frac{1}{17} \quad \frac{1}{26}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

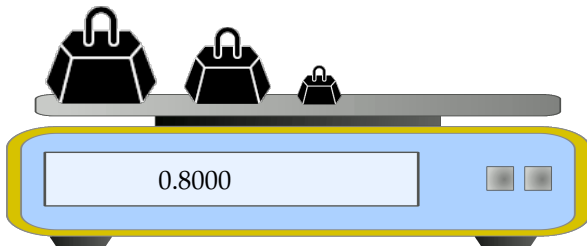
CONVERGES

$$S_1 = 0.5000$$

$$S_2 = 0.7000$$

$$S_3 = 0.8000$$

$$\frac{1}{17} \quad \frac{1}{26}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

CONVERGES

$$S_1 = 0.5000$$

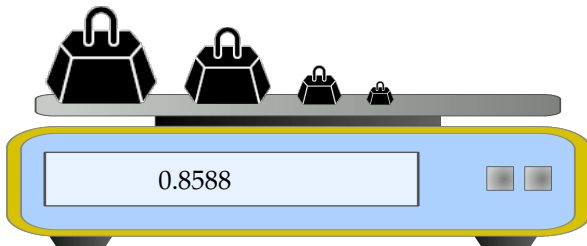
$$S_2 = 0.7000$$

$$S_3 = 0.8000$$

$$S_4 = 0.8588$$



$$\frac{1}{26}$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$$

CONVERGES

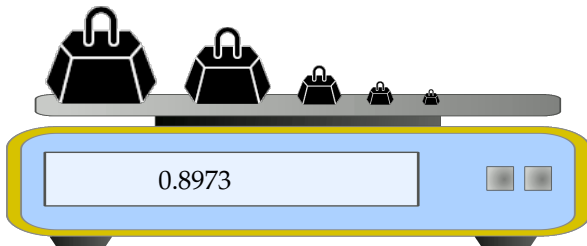
$$S_1 = 0.5000$$

$$S_2 = 0.7000$$

$$S_3 = 0.8000$$

$$S_4 = 0.8588$$

$$S_5 = 0.8973$$

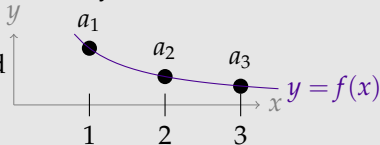


Integral Test

Let N_0 be any natural number. If $f(x)$ is a function which is defined and continuous for all $x \geq N_0$ and which obeys

- (i) $f(x) \geq 0$ for all $x \geq N_0$ and
- (ii) $f(x)$ decreases as x increases and
- (iii) $f(n) = a_n$ for all $n \geq N_0$.

Then



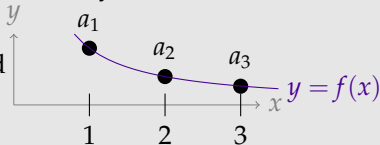
$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_{N_0}^{\infty} f(x) \, dx \text{ converges}$$

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$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_{N_0}^{\infty} f(x) \, dx \text{ converges}$$

Furthermore, when the series converges, the truncation error satisfies

$$0 \leq \sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n \leq \int_N^{\infty} f(x) \, dx \quad \text{for all } N \geq N_0$$

Does the series $\sum_{n=10}^{\infty} \frac{1}{n \log n}$ converge or diverge?

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If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=a}^{\infty} a_n$ diverges.

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No use here: we need another test.

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Divergence Test

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=a}^{\infty} a_n$ diverges.

No use here: we need another test.

Set $f(x) = \frac{1}{x \log x}$.

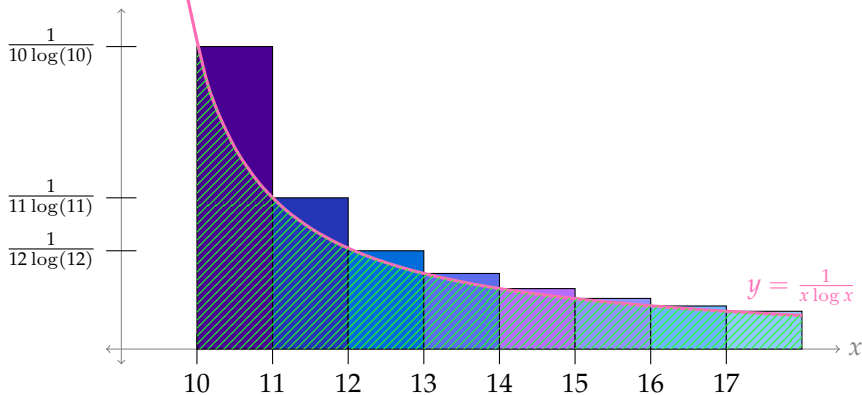
- (i) $f(x) \geq 0$ for all $x \geq 10$ and
- (ii) $f(x)$ decreases as x increases and
- (iii) $f(n) = a_n$ for all $n \geq 10$.

So, the integral test applies.

Does the series $\sum_{n=10}^{\infty} \frac{1}{n \log n}$ converge or diverge?

$$\int_{10}^{\infty} \frac{1}{x \log x} dx =$$

Does the series $\sum_{n=10}^{\infty} \frac{1}{n \log n}$ converge or diverge?

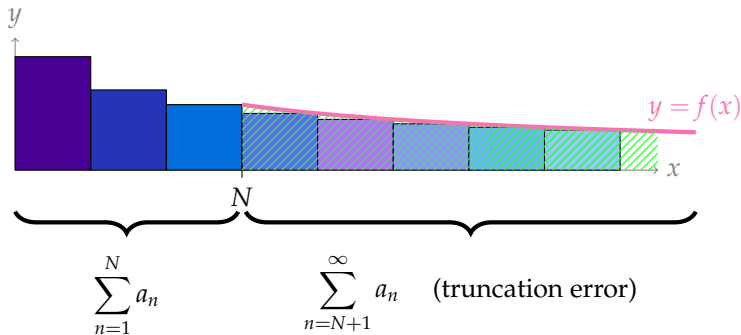


$$\int_{10}^{\infty} \frac{1}{x \log x} dx = \infty$$

Integral Test, abridged

... When the series converges, the truncation error satisfies

$$0 \leq \sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n \leq \int_N^{\infty} f(x) \, dx$$



Integral Test, abridged

When the series converges, the truncation error satisfies

$$0 \leq \sum_{n=1}^{\infty} a_n - \sum_{n=1}^N a_n \leq \int_N^{\infty} f(x) \, dx$$

We already decided that the series $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$ converges.

Suppose we had a computer add up the terms $n = 1$ through $n = 100$.

Use the integral test to bound the error, $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1} - \sum_{n=1}^{100} \frac{1}{n^2 + 1}$.

By computer, $\sum_{n=1}^{100} \frac{1}{n^2 + 1} \approx 1.0667$. Using the truncation error of about

0.01, give a (small) range of possible values for $\sum_{n=1}^{\infty} \frac{1}{n^2 + 1}$.

$$0 \leq \sum_{n=1}^{\infty} \frac{1}{n^2 + 1} - \sum_{n=1}^{100} \frac{1}{n^2 + 1} \leq \int_{100}^{\infty} \frac{1}{x^2 + 1} dx$$



p -TEST

Let p be a positive constant. When we talked about improper integrals, we showed:

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \begin{cases} \text{converges if } p > 1 \\ \text{diverges if } p \leq 1 \end{cases}$$

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$$\int_1^{\infty} \frac{1}{x^p} dx \quad \begin{cases} \text{converges if } p > 1 \\ \text{diverges if } p \leq 1 \end{cases}$$

Set $f(x) = \frac{1}{x^p}$.

- (i) $f(x) \geq 0$ for all $x \geq 1$, and
- (ii) $f(x)$ decreases as x increases

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \left\{ \right.$$

Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^3}.$$

By the p -test, we know this series

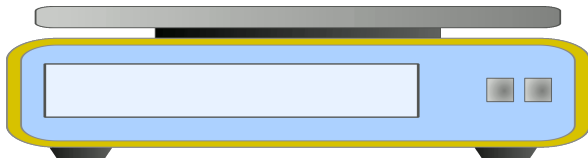
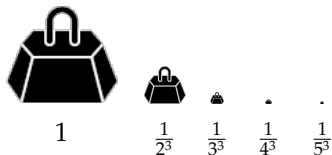
Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{n^3}.$$

By the p -test, we know this series converges.

How many terms should we add up to approximate the series to within an error of no more than 0.02?

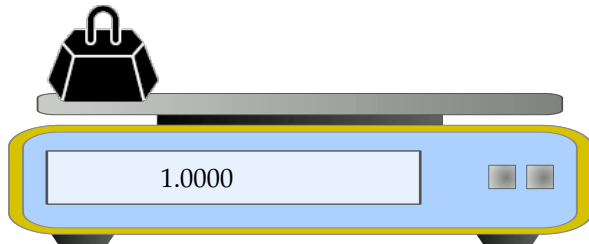
$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.



$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.

$$S_1 = 1.0000$$

			
$\frac{1}{2^3}$	$\frac{1}{3^3}$	$\frac{1}{4^3}$	$\frac{1}{5^3}$

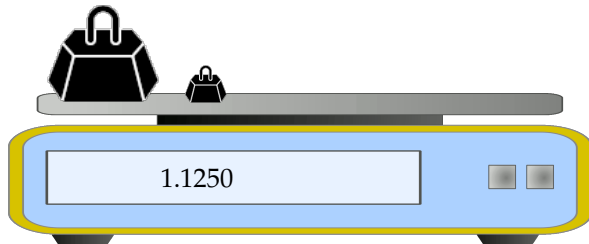


$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.

$$S_1 = 1.0000$$

$$S_2 = 1.1250$$

$$\begin{array}{ccc} \text{⬆} & \text{⬆} & \text{⬆} \\ \frac{1}{3^3} & \frac{1}{4^3} & \frac{1}{5^3} \end{array}$$



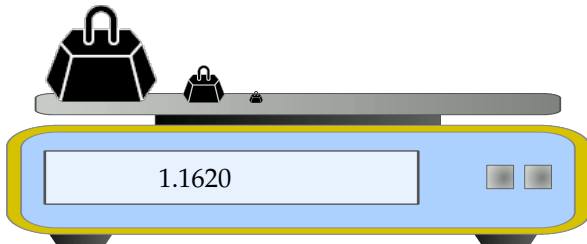
$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.

$$S_1 = 1.0000$$

$$S_2 = 1.1250$$

$$S_3 = 1.1620$$

$$\frac{1}{4^3} \quad \frac{1}{5^3}$$



$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.

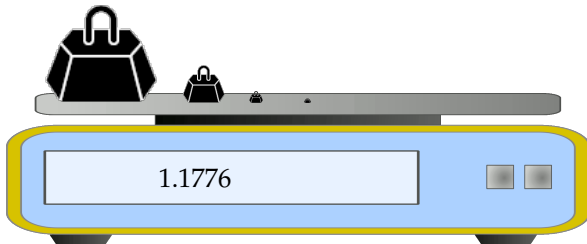
$$S_1 = 1.0000$$

$$S_2 = 1.1250$$

$$S_3 = 1.1620$$

$$S_4 = 1.1776$$

$$\frac{1}{5^3}$$



$\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges to within 0.02 of $\sum_{n=1}^5 \frac{1}{n^3}$.

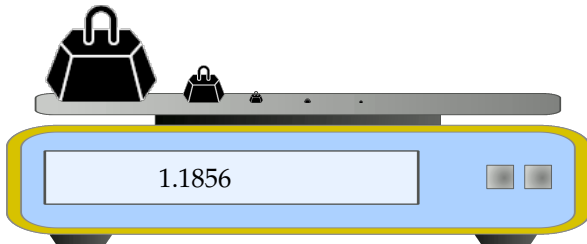
$$S_1 = 1.0000$$

$$S_2 = 1.1250$$

$$S_3 = 1.1620$$

$$S_4 = 1.1776$$

$$S_5 = 1.1856$$



Observation

In a series with **positive** terms, the series either **converges**, or **diverges to infinity**.

If terms are “too big,” series will diverge.

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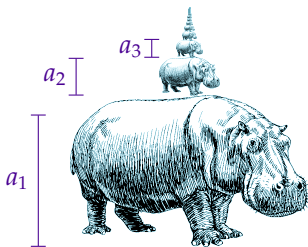
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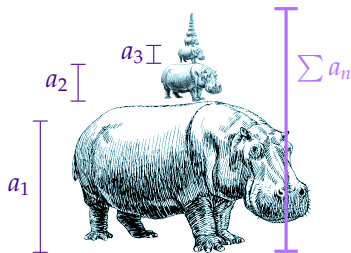
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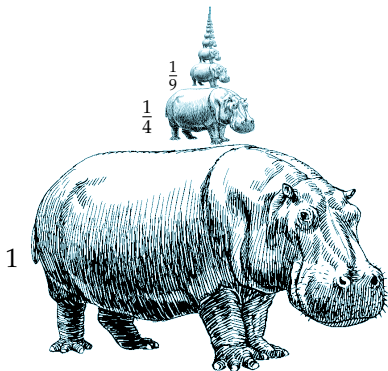


$\sum \frac{1}{n^2}$ converges

$$\sum \frac{1}{n^2 + n}$$

$$\sum \frac{1}{n^2} \text{ converges}$$

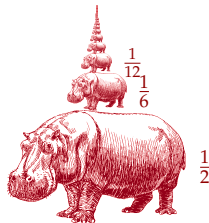
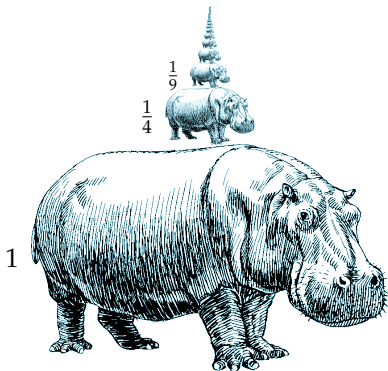
$$\sum \frac{1}{n^2 + n}$$



Terms are “small enough” for
sum to converge

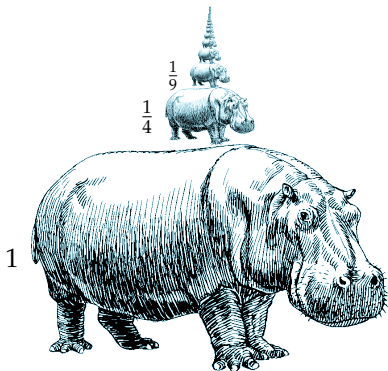
$$\sum \frac{1}{n^2} \text{ converges}$$

$$\sum \frac{1}{n^2 + n}$$



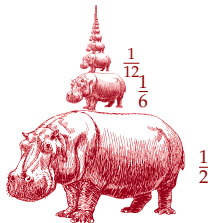
Terms are “small enough” for
sum to converge

$$\sum \frac{1}{n^2} \text{ converges}$$



Terms are “small enough” for
sum to converge

$$\sum \frac{1}{n^2 + n} \text{ converges, too}$$



Terms are also “small enough”
for sum to converge

The Comparison Test

Let N_0 be a natural number and let $K > 0$.

- (a) If $|a_n| \leq Kc_n$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} c_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges.
- (b) If $a_n \geq Kd_n \geq 0$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} d_n$ diverges, then $\sum_{n=0}^{\infty} a_n$ diverges.

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Consider $\sum_{n=1}^{\infty} \frac{1}{n^{0.1}}$.

The Comparison Test

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- We know $0 < \frac{1}{n} < \frac{1}{n-0.1}$

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Consider $\sum_{n=1}^{\infty} \frac{1}{n-0.1}$.

- ▶ We know $0 < \frac{1}{n} < \frac{1}{n-0.1}$
- ▶ We know $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (harmonic series)

The Comparison Test

Let N_0 be a natural number and let $K > 0$.

- (a) If $|a_n| \leq Kc_n$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} c_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges.
- (b) If $a_n \geq Kd_n \geq 0$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} d_n$ diverges, then $\sum_{n=0}^{\infty} a_n$ diverges.

Consider $\sum_{n=1}^{\infty} \frac{1}{n-0.1}$.

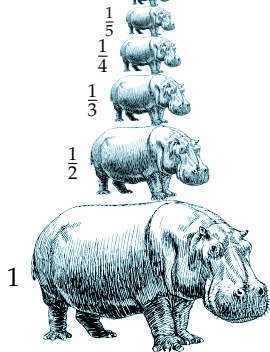
- ▶ We know $0 < \frac{1}{n} < \frac{1}{n-0.1}$
- ▶ We know $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges (harmonic series)
- ▶ So, by the comparison test, $\sum_{n=1}^{\infty} \frac{1}{n-0.1}$ diverges as well.

$$\sum \frac{1}{n} \text{ diverges}$$

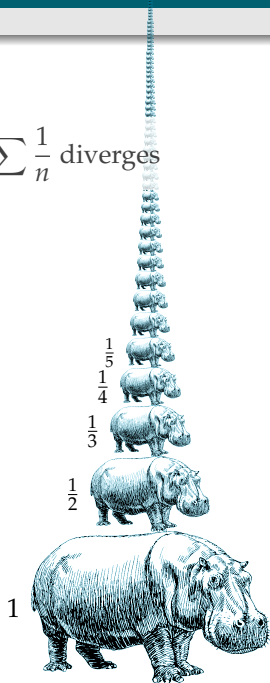
$$\sum \frac{1}{n - 0.1}$$

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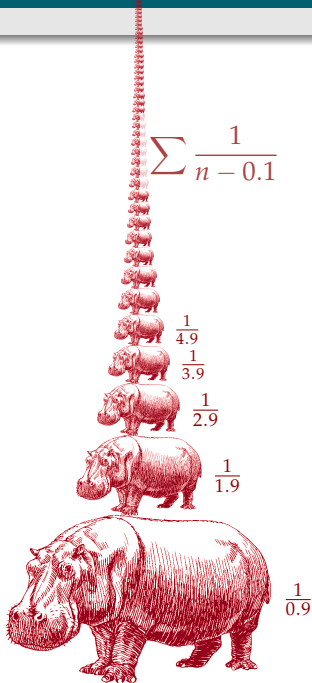
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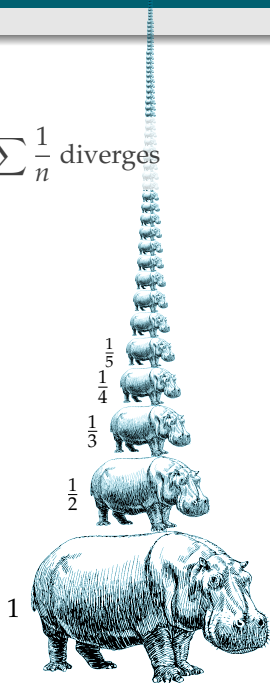
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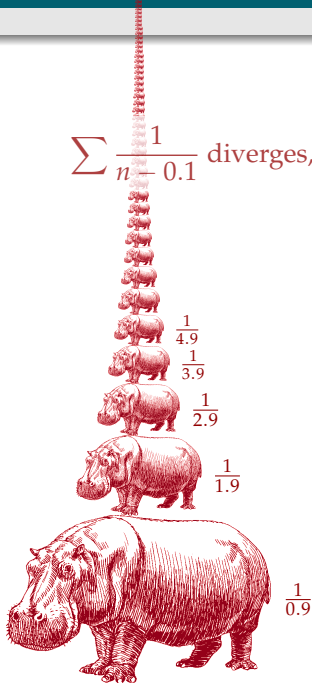
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$$\sum \frac{1}{n} \text{ diverges}$$



$$\sum \frac{1}{n - 0.1} \text{ diverges, too}$$



Does the series $\sum_{n=1}^{\infty} \frac{n + \cos n}{n^3 - 1/3}$ converge or diverge?

Step 1: Intuition.

When n is very large, we expect:

► $n + \cos n \approx$

► $n^3 + \frac{1}{3} \approx$

► So, we expect $\frac{n + \cos n}{n^3 - 1/3} \approx$

Since $\sum_{n=1}^{\infty} \frac{1}{n^2} \dots$

we expect $\sum_{n=1}^{\infty} \frac{n + \cos n}{n^3 - 1/3}$ to also

Does the series $\sum_{n=1}^{\infty} \frac{n + \cos n}{n^3 - 1/3}$ converge or diverge?

Step 2: Choose comparison series.

The Comparison Test, abridged

Let N_0 be a natural number and let $K > 0$.

If $|a_n| \leq Kc_n$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} c_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges.

To show that original series **converges**, we should find a comparison series that also **converges** and whose terms (times some positive constant) are **larger** than the original terms. *There are many possibilities.* For $n \geq 1$,

- ▶ $n + \cos n <$
- ▶ $n^3 - \frac{1}{3} >$
- ▶ So $\frac{n + \cos n}{n^3 - 1/3} <$

Does the series $\sum_{n=1}^{\infty} \frac{n + \cos n}{n^3 - 1/3}$ converge or diverge?

Step 3: Verify.

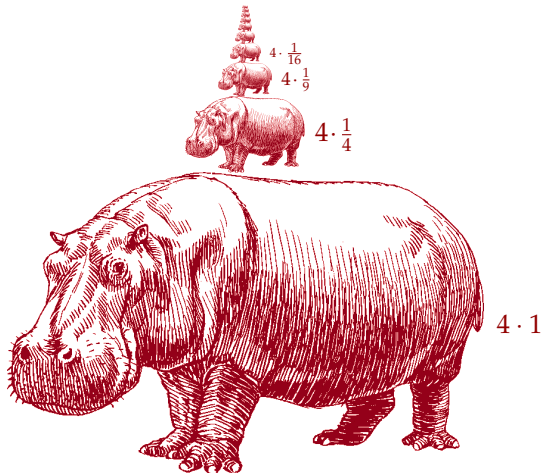
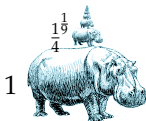
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Let N_0 be a natural number and let $K > 0$.

If $|a_n| \leq Kc_n$ for all $n \geq N_0$ and $\sum_{n=0}^{\infty} c_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges.

$\sum \frac{1}{n^2}$ converges, so

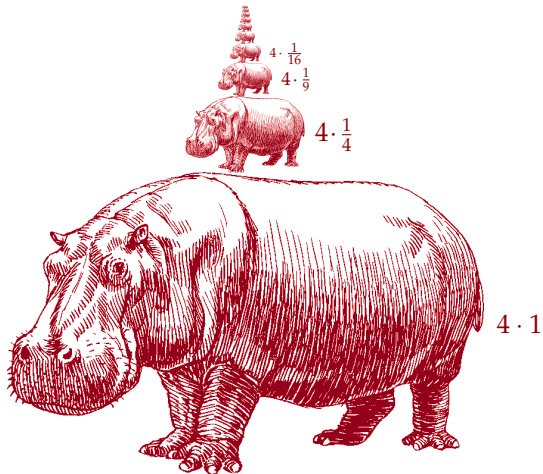
$\sum 4 \cdot \frac{1}{n^2}$ converges, too



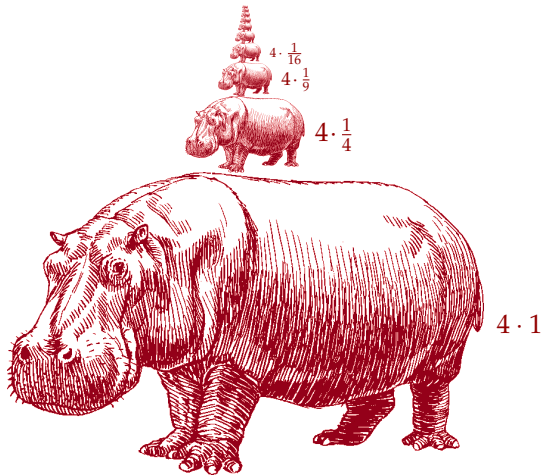
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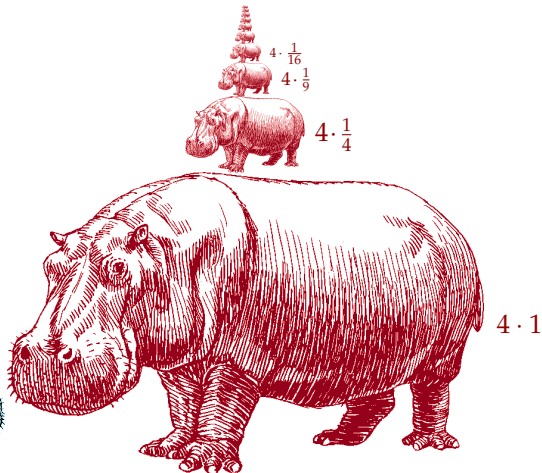
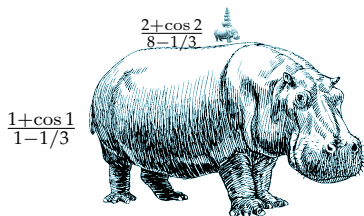


$$\sum 4 \cdot \frac{1}{n^2} \text{ converges}$$

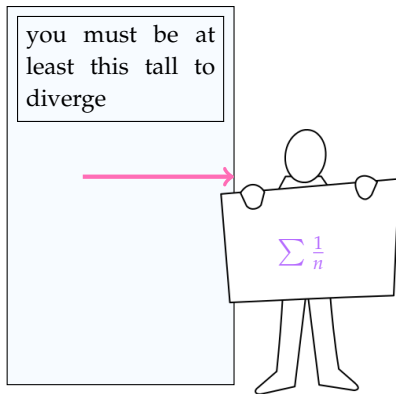


So, $\sum \frac{n + \cos n}{n^3 - 1/3}$ converges too.

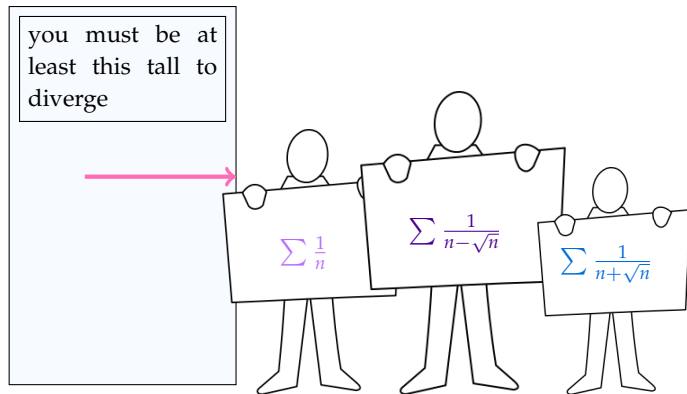
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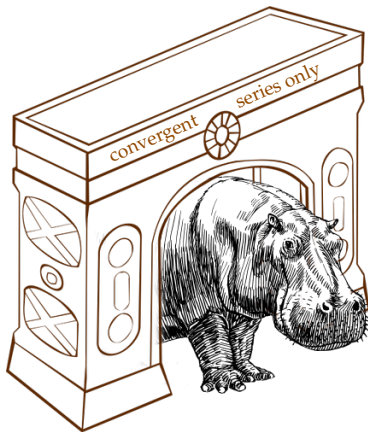
For the comparison test as we have seen it so far, to conclude that a given series **diverges**, we have to find a divergent comparison series whose terms are **smaller** than (a positive multiple of) those of our original series .



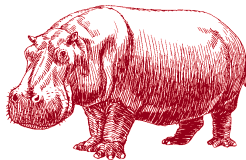
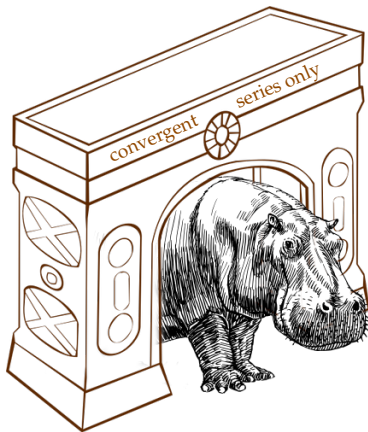
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Limit Comparison Theorem

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series with $b_n > 0$ for all n . Assume that

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$$

exists.

(a) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges too.

(b) If $L \neq 0$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges too.

In particular, if $L \neq 0$, then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.

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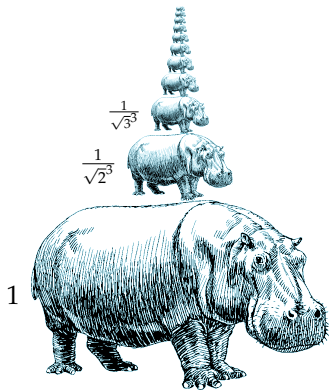
In particular, if $L \neq 0$, then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.

- ▶ For large n , $a_n \approx L \cdot b_n$;
- ▶ so we expect $\sum a_n$ to behave roughly like $\sum (L \cdot b_n)$;
- ▶ and since $L \neq 0$, we expect $\sum (L \cdot b_n)$ to converge if and only if $\sum b_n$ converges.

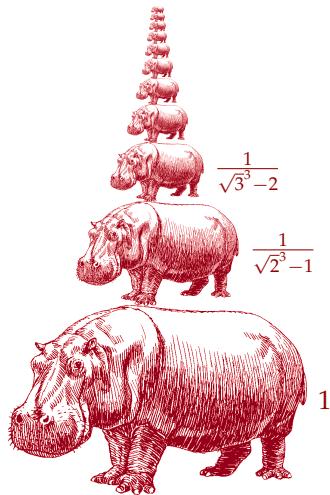
By the p -test, $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges.

Can we conclude that $\sum_{n=1}^{\infty} \frac{1}{n^{3/2} - n + 1}$ also converges?

$\sum \frac{1}{n^{3/2}}$ converges.



So, $\sum \frac{1}{n^{3/2} - n + 1}$ converges too.



Does the series $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}}{n^2 - 2n + 3}$ converge or diverge?

Step 1: Intuition

For large n ,

Does the series $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}}{n^2 - 2n + 3}$ converge or diverge?

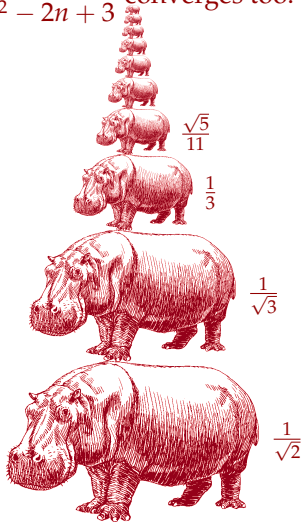
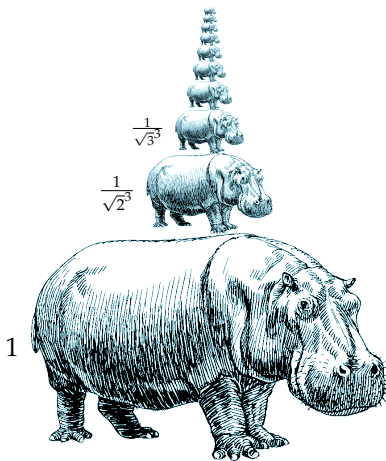
Step 2: Verify Intuition

Let $a_n = \frac{\sqrt{n+1}}{n^2 - 2n + 3}$ and $b_n = \frac{1}{n^{3/2}}$.

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} =$$

$\sum \frac{1}{n^{3/2}}$ converges.

So, $\sum \frac{\sqrt{n+1}}{n^2 - 2n + 3}$ converges too.



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- ▶ Common guess (especially if monotone): consider “largest” piece of numerator and denominator
(constant) < (logarithm) < (polynomial) < (exponential)
- ▶ After you guess a comparison series, **show it works** by finding the correct inequality (comparison test), or computing the limit of the ratio (limit comparison test).

CHOOSE A SERIES TO COMPARE

$$\sum_{n=1}^{\infty} \frac{3n}{n^2 + 1}$$

$$\sum_{n=1}^{\infty} \frac{n^2 + n + 1}{n^5 - n}$$

$$\sum_{k=1}^{\infty} \frac{k(2 + \sin k)}{k^{\sqrt{2}}}$$

$$\sum_{m=1}^{\infty} \frac{3m + \sin \sqrt{m}}{m^2}$$

Included Work

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